

LIPERI TUISKAVANLUOTO, Finland

WMO: 027930

Lat: 62.546N

Lon: 29.668E

Elev: 80

StdP: 100.37

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.9	(c) -23.5	(d) -29.8	(e) 0.2	(f) -26.9	(g) -26.0	(h) 0.4	(i) -23.5	(j) 13.2	(k) -5.6	(l) 12.3	(m) -4.4	(n) 1.2	(o) 0	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.4	25.8	20.3	24.1	19.3	22.5	18.5	21.3	24.4	20.2	23.2	18.9	22.1	3.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.2	15.0	23.5	18.9	13.9	22.4	17.5	12.7	21.0	62.4	24.4	58.2	23.3	54.1	22.0	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
12.2	11.1	10.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-27.9	26.8	5.4	2.5	-31.9	28.6	-35.0	30.1	-38.1	31.5	-42.0	33.3
			-28.3	21.4	5.1	2.5	-32.0	23.2	-35.0	24.7	-37.8	26.1	-41.5	27.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.3	-9.1	-7.4	-4.7	1.1	9.2	14.4	18.3	16.5	11.7	4.9	0.2	-4.0
	DBStd	10.38	7.46	7.15	5.29	3.31	4.54	3.77	3.52	3.10	3.14	3.22	3.79	6.24
	HDD10.0	2784	593	487	456	266	70	6	0	1	20	159	293	434
	HDD18.3	5182	851	720	714	516	283	128	46	74	199	416	543	692
	CDD10.0	716	0	0	0	0	47	138	257	202	71	1	0	0
	CDD18.3	71	0	0	0	0	1	9	44	17	0	0	0	0
	CDH23.3	232	0	0	0	0	1	33	160	38	0	0	0	0
	CDH26.7	24	0	0	0	0	0	4	18	2	0	0	0	0
	WSAvg	5.2	5.1	5.0	5.2	4.4	4.7	5.0	4.6	4.9	5.6	6.2	6.1	5.7
	PrecAvg	619	47	38	33	30	41	62	69	73	51	63	56	52
(15) Precipitation	PrecMax	729	79	66	71	67	76	123	134	188	140	127	130	81
	PrecMin	532	16	5	8	5	12	33	28	27	20	13	15	20
	PrecStd	69	18	19	13	15	19	25	27	40	28	30	26	18
	PrecStd	69	18	19	13	15	19	25	27	40	28	30	26	18
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	2.6	3.7	5.2	11.3	22.6	26.7	28.7	26.2	19.6	12.1	8.5	4.9	
	0.4% MCWB	1.9	2.2	2.4	6.5	15.6	19.4	21.4	20.8	16.2	11.0	7.5	4.4	
	2% DB	1.6	2.1	3.5	8.4	20.1	23.8	26.8	24.4	17.7	11.0	6.7	3.3	
	2% MCWB	1.0	1.3	1.5	4.9	14.7	18.1	20.8	19.5	15.3	9.9	5.9	2.6	
	5% DB	0.9	1.2	2.5	6.5	17.7	21.4	25.1	22.5	16.6	10.1	5.8	2.3	
	5% MCWB	0.5	0.8	1.1	3.8	13.4	17.4	20.2	18.5	14.7	9.0	5.3	1.8	
	10% DB	0.2	0.6	1.6	5.1	15.7	19.6	23.6	20.9	15.8	9.2	4.5	1.6	
	10% MCWB	-0.1	0.3	0.6	2.8	11.9	15.7	19.4	17.6	14.0	8.1	4.0	1.1	
	0.4% WB	2.2	2.3	2.8	7.2	16.8	21.1	22.8	22.4	17.4	11.6	7.5	4.4	
	0.4% MCDB	2.5	3.0	4.0	10.1	20.7	24.2	26.5	25.2	18.4	11.9	8.1	4.7	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	1.2	1.5	2.0	5.4	15.4	19.1	21.8	20.1	16.2	10.4	6.1	2.8	
	2% MCDB	1.5	2.0	2.9	7.9	18.9	22.7	25.1	23.0	17.1	10.8	6.6	3.1	
	5% WB	0.6	0.9	1.4	4.1	13.9	17.6	21.0	19.0	15.1	9.1	5.4	1.8	
	5% MCDB	0.9	1.2	2.1	6.1	17.5	20.9	23.9	21.9	16.2	9.8	5.8	2.2	
	10% WB	0.0	0.3	0.8	3.1	12.2	16.1	20.0	17.9	14.2	8.2	4.1	1.2	
	10% MCDB	0.3	0.6	1.5	4.6	15.3	19.1	23.2	20.5	15.4	9.0	4.4	1.5	
(35) Mean Daily Temperature Range	MDBR	6.0	6.1	7.6	6.5	6.3	5.5	5.4	4.5	3.7	3.0	2.7	3.7	
	5% DB	3.2	4.4	6.5	8.3	9.0	7.8	7.1	6.1	4.6	3.4	3.4	2.7	
	5% WB	3.0	4.2	5.4	5.2	4.8	4.4	3.8	3.4	3.1	3.0	3.1	2.6	
	MCDBR	3.1	4.0	5.4	7.1	8.5	7.4	6.5	5.3	4.0	3.4	3.3	2.7	
	MCWBR	3.0	3.7	4.7	4.7	5.0	4.6	4.1	3.3	3.2	3.3	3.2	2.7	
	MCWBR	3.0	3.7	4.7	4.7	5.0	4.6	4.1	3.3	3.2	3.3	3.2	2.7	
(40) Clear-Sky Solar Irradiance	taub	0.235	0.288	0.315	0.354	0.353	0.348	0.374	0.357	0.335	0.314	0.252	0.214	
	taud	2.009	2.107	2.178	2.315	2.446	2.489	2.431	2.462	2.511	2.397	2.103	1.973	
	Ebn at Noon	407	642	779	829	866	877	839	824	775	634	385	235	
	Edh at Noon	35	69	94	102	98	96	99	88	70	53	31	19	
(44) All-Sky Solar Radiation	RadAvg	0.20	0.77	2.29	3.67	4.83	5.31	5.06	3.86	2.14	0.82	0.21	0.09	
	RadStd	0.03	0.18	0.31	0.44	0.55	0.30	0.48	0.45	0.25	0.13	0.04	0.01	

Historical Trends

	DBAvg	Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (28 neighbors)	+0.55	N/A	N/A	N/A	N/A	N/A	-191	N/A	N/A

Nomenclature: See separate page